

MLWP VIVA QUESTIONS BASE ON ASSIGNMENTS

Assignment 1

🎯 1. What is Machine Learning?

Answer:

Machine Learning is a branch of artificial intelligence that allows systems to learn from data and improve performance without being explicitly programmed. Instead of writing rules, the model learns patterns from data automatically.

🎯 2. Why is Machine Learning important?

Answer:

Machine Learning is important because it automates decision-making, handles large datasets efficiently, and provides accurate predictions. It also improves over time as more data is available.

🎯 3. Give some real-world applications of Machine Learning.

Answer:

Some common applications are:

- Spam email detection
 - Recommendation systems like Netflix and Amazon
 - Fraud detection in banking
 - Face recognition
 - Self-driving cars
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🎯 4. What is Reinforcement Learning?

Answer:

Reinforcement Learning is a type of machine learning where an agent learns by interacting with the environment and receiving rewards or penalties. The goal is to maximize the total reward over time.

🎯 5. What are the main components of Reinforcement Learning?



Answer:

The main components are:

- Agent – decision maker
 - Environment – where agent works
 - Action – what agent does
 - Reward – feedback from environment
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🎯 6. Explain the working of Reinforcement Learning.**Answer:**

First, the agent takes an action. The environment responds and gives a reward or penalty. Based on this feedback, the agent updates its strategy. This process repeats until the agent learns the best actions.

🎯 7. Why is spam detection a supervised learning problem?**Answer:**

Because it uses labeled data. Emails are already marked as spam or not spam, so the model learns from these examples. That makes it a classification problem under supervised learning.

🎯 8. How is a spam detection model trained?**Answer:**

First, data is collected and labeled. Then preprocessing is done to clean and convert text into features. After that, a model is trained and evaluated. Finally, it is used to classify new emails.

🎯 9. What is the role of evaluation in ML models?**Answer:**

Evaluation checks how well the model performs on unseen data. Common metrics include accuracy, precision, recall, and confusion matrix.

🎯 10. What is a scatter plot in ML?**Answer:**

A scatter plot is a visualization technique used to show the relationship between two numerical features. Each point represents a data sample.



🔗 11. What is the use of “hue” in Seaborn?

Answer:

Hue is used to represent different categories using colors. For example, in the Iris dataset, different species are shown using different colors.

🔗 12. What can you observe from the Iris scatter plot?

Answer:

Setosa is clearly separated, while Versicolor and Virginica overlap. This shows that some classes are easy to classify while others are not.

🔗 13. How does ML help in recommendation systems?

Answer:

Machine Learning analyzes user data like history, ratings, and preferences. It learns patterns and suggests relevant items like movies or products.

🔗 14. Give example of recommendation system.

Answer:

Netflix recommends movies based on viewing history, and Amazon suggests products based on past purchases and searches.

🔗 15. What are benefits of recommendation systems?

Answer:

- Personalized experience
 - Increased user engagement
 - Better sales and retention
 - Time-saving for users
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Assignment 2

🔗 1. What is Clustering?

Answer:

Clustering is an unsupervised learning technique where similar data points are grouped together into



clusters, such that points in the same cluster are more similar and points in different clusters are less similar.

🎯 2. Is clustering supervised or unsupervised? Why?

Answer:

Clustering is unsupervised learning because it does not use labeled data. The algorithm groups data automatically based on similarity.

🎯 3. What is the goal of clustering?

Answer:

The goal is to achieve high similarity within clusters and low similarity between clusters.

🎯 4. What are distance metrics in clustering?

Answer:

Distance metrics are used to measure similarity between data points. Common ones include Euclidean distance and Manhattan distance.

🎯 5. Difference between Euclidean and Manhattan distance?

Answer (short viva style):

Euclidean distance is the straight-line distance between two points (L2 norm), while Manhattan distance is the sum of absolute differences along axes (L1 norm).

🎯 6. What is K-Means clustering?

Answer:

K-Means is a clustering algorithm that divides data into K clusters by assigning points to the nearest centroid and updating centroids iteratively.

🎯 7. What type of clusters does K-Means form?

Answer:

K-Means forms spherical or circular clusters and works best when clusters are compact and well-separated.

🎯 8. Limitations of K-Means?



Answer:

- Requires predefined K
 - Sensitive to outliers
 - Not suitable for non-spherical shapes
 - Sensitive to initial centroid selection
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9. What is DBSCAN?

Answer:

DBSCAN is a density-based clustering algorithm that groups points based on density and can detect clusters of arbitrary shapes.

10. Advantage of DBSCAN over K-Means?

Answer:

DBSCAN can detect clusters of any shape and also identifies noise (outliers), unlike K-Means.

11. Limitations of DBSCAN?

Answer:

- Difficult to choose parameters
 - Not good for varying density clusters
 - Performance decreases in high dimensions
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12. What is PCA?

Answer:

PCA (Principal Component Analysis) is a dimensionality reduction technique that converts original features into new uncorrelated variables called principal components while preserving maximum variance.

13. Why do we use PCA?

Answer:

- Reduce number of features
 - Remove redundancy
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- Improve visualization
 - Speed up algorithms
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🔗 14. What are principal components?

Answer:

Principal components are new variables formed as combinations of original features that capture maximum variance and are uncorrelated.

🔗 15. What is SSE in K-Means?

Answer:

SSE (Sum of Squared Error) is the sum of squared distances between each data point and its cluster centroid. It measures clustering quality (lower is better).

Assignment 3

1. What is Machine Learning?

Answer:

Machine Learning is a branch of AI that enables systems to learn from data and improve performance without being explicitly programmed.

2. What are types of Machine Learning?

Answer:

- Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
-

3. What is Reinforcement Learning?

Answer:

It is a type of learning where an agent interacts with an environment and learns by receiving rewards or penalties, aiming to maximize total reward.

4. What is Clustering?



Answer:

Clustering is an unsupervised technique where similar data points are grouped together without using labeled data.

5. What is the difference between supervised and unsupervised learning?

Answer:

Supervised learning uses labeled data, while unsupervised learning works with unlabeled data and finds patterns automatically.

6. What is K-Means clustering?

Answer:

K-Means is an algorithm that partitions data into K clusters by assigning points to the nearest centroid and updating centroids iteratively.

7. What is DBSCAN?

Answer:

DBSCAN is a density-based clustering algorithm that forms clusters based on density and can detect noise points.

8. What is PCA?

Answer:

PCA is a dimensionality reduction technique that converts original features into new uncorrelated components while preserving maximum variance.

9. What is Feature Engineering?

Answer:

Feature engineering is the process of transforming raw data into meaningful features to improve model performance.

10. Why is Feature Engineering important?

Answer:

It improves model accuracy, simplifies learning, handles complex relationships, and improves data quality.



11. What is Feature Scaling?

Answer:

Feature scaling is a preprocessing technique used to bring all feature values into a similar range so that no feature dominates others.

12. What is Normalization?

Answer:

Normalization is a type of scaling where data is transformed into a fixed range, usually between 0 and 1.

13. How do you handle missing values in a dataset?

Answer:

- Deletion (row/column removal)
 - Mean/Median/Mode imputation
 - KNN imputation
 - Model-based imputation
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14. What are feature selection methods?

Answer:

- Filter method (statistical, fast)
 - Wrapper method (model-based, accurate but slow)
 - Embedded method (done during training)
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15. What is an Outlier?

Answer:

An outlier is a data point that is significantly different from other values and can affect model performance.

Assignment 4

1. What is Machine Learning?



Answer:

Machine Learning is a branch of AI where systems learn patterns from data and improve performance without explicit programming.

2. What is the difference between supervised and unsupervised learning?

Answer:

Supervised learning uses labeled data, while unsupervised learning works with unlabeled data and finds patterns automatically.

3. What is Reinforcement Learning?

Answer:

It is a learning method where an agent interacts with an environment and learns using rewards and penalties to maximize performance.

4. What is Clustering?

Answer:

Clustering is an unsupervised technique that groups similar data points together based on similarity.

5. What is K-Means clustering?

Answer:

K-Means is a clustering algorithm that divides data into K groups using centroids and updates them iteratively.

6. What is DBSCAN?

Answer:

DBSCAN is a density-based clustering algorithm that forms clusters based on data density and can detect noise points.

7. What is PCA?

Answer:

PCA is a dimensionality reduction technique that transforms data into new uncorrelated variables while preserving maximum variance.

8. What is Feature Engineering?



Answer:

Feature engineering is the process of transforming raw data into meaningful features to improve model accuracy.

9. What is Feature Scaling?

Answer:

Feature scaling adjusts feature values into a similar range so that all features contribute equally to the model.

10. What is Normalization?

Answer:

Normalization is a scaling technique that converts data into a fixed range, usually between 0 and 1.

11. How do you handle missing data?

Answer:

Missing data can be handled using deletion methods, mean/median/mode imputation, KNN imputation, or model-based techniques.

12. What is Overfitting and Underfitting?

Answer:

Overfitting occurs when a model learns training data too well but fails on new data, while underfitting happens when the model is too simple to learn patterns properly.

13. What is ROC Curve?

Answer:

ROC Curve is a graph that shows the performance of a classification model by plotting True Positive Rate against False Positive Rate at different thresholds.

14. What is AUC?

Answer:

AUC (Area Under Curve) measures the overall performance of a model. Higher AUC indicates better classification ability.

15. What is K-Fold Cross Validation?



Answer:

It is a technique where data is divided into K parts, and the model is trained and tested multiple times to get reliable performance.

Assignment 5

1. What is Machine Learning?

Answer:

Machine Learning is a branch of AI where systems learn patterns from data and improve performance without explicit programming.

2. What are types of Machine Learning?

Answer:

- Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
-

3. What is Reinforcement Learning?

Answer:

It is a learning technique where an agent learns by interacting with the environment using rewards and penalties.

4. What is Clustering?

Answer:

Clustering is an unsupervised technique that groups similar data points together based on similarity.

5. What is K-Means clustering?

Answer:

K-Means is a clustering algorithm that divides data into K clusters by assigning points to nearest centroids and updating them iteratively.

6. What is PCA?



Answer:

PCA is a dimensionality reduction technique that converts data into new uncorrelated features while preserving maximum variance.

7. What is Feature Engineering?

Answer:

Feature engineering is the process of transforming raw data into meaningful features to improve model performance.

8. What is Feature Scaling?

Answer:

Feature scaling adjusts feature values into a similar range so that all features contribute equally to the model.

9. What is Overfitting and Underfitting?

Answer:

Overfitting occurs when a model performs well on training data but poorly on new data, while underfitting occurs when the model fails to learn patterns.

10. What is ROC Curve?

Answer:

ROC Curve is a graphical representation that shows model performance by plotting True Positive Rate against False Positive Rate.

11. What is K-Fold Cross Validation?

Answer:

It is a technique where data is divided into K parts, and the model is trained and tested multiple times for reliable evaluation.

12. What is Bagging?

Answer:

Bagging is an ensemble technique where multiple models are trained on different random samples and their outputs are combined to improve performance.



13. What is Random Forest?

Answer:

Random Forest is an ensemble method that improves bagging by using random feature selection along with multiple decision trees, increasing accuracy and reducing overfitting.

14. What is Voting Ensemble?

Answer:

Voting combines predictions from multiple models. Hard voting uses majority vote, while soft voting uses probability averaging.

15. What is Stacking?

Answer:

Stacking is an ensemble technique where multiple models are combined using a meta-model that learns how to best combine their predictions.

